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## Calculation (saddle points and nodes)

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On the last page, some new notation was introduced:

$$\vec{M}(t) = \vec{X}(t) - \vec{X}_0 = \begin{bmatrix} P(t) \\ G(t) \end{bmatrix} - \begin{bmatrix} P_0 \\ G_0 \end{bmatrix},$$

where  $(P_0, G_0)$  is an equilibrium point. You have learned that a system of differential equations that is linearised around an equilibrium point, can be written as:

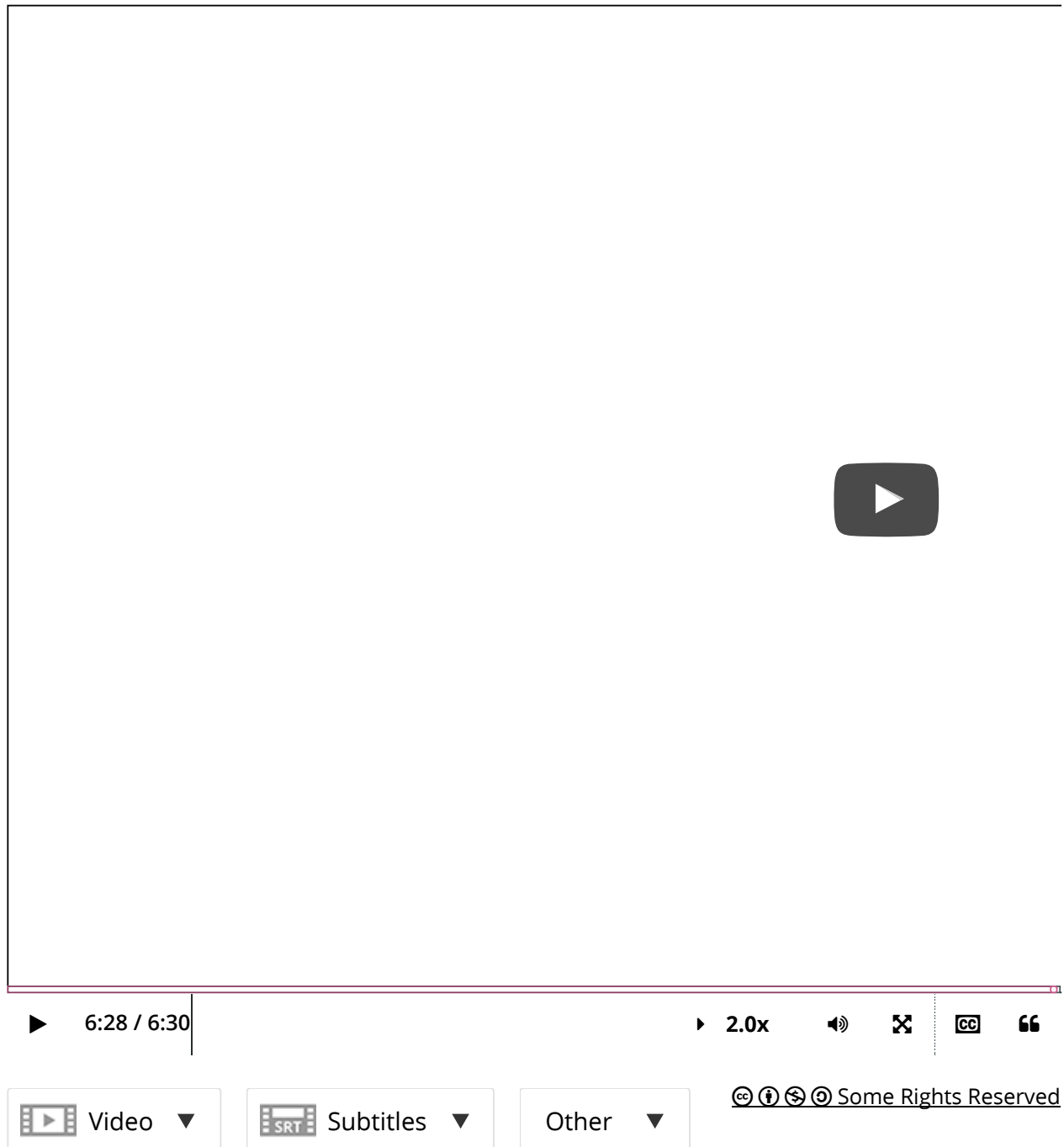
$$\frac{d\vec{M}}{dt} = J(\vec{X}_0) \vec{M}(t).$$

When our rainbowfish/gourami system is linearised around  $(P_0, G_0) = (100, 0)$ , the result is:

$$\frac{d\vec{M}}{dt} = \begin{bmatrix} -0.7 & -4 \\ 0 & 0.55 \end{bmatrix} \vec{M}(t).$$

In the next video, Dennis explains what this new differential equation can tell you about the original differential equations.

## Equilibrium points in a system



6:28 / 6:30

2.0x

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Video Subtitles Other

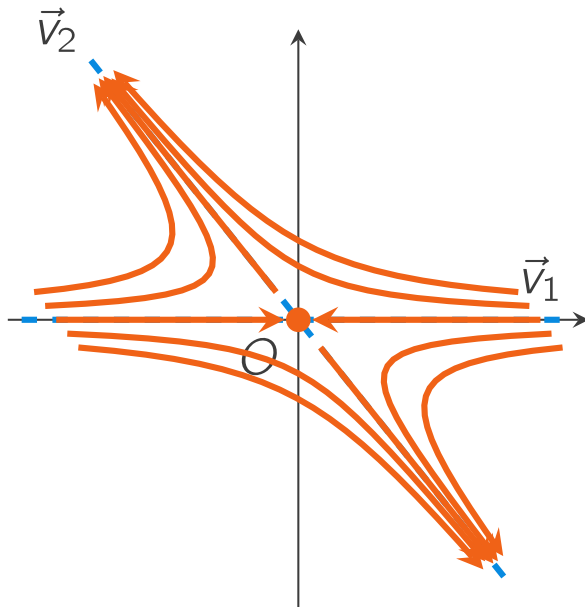
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In the video you have seen that  $(100, 0)$  is a saddle point by looking at the solutions of the linearised differential equation. The behaviour of solutions near a saddle point is explained by the eigenvalues of the Jacobian matrix: one is positive, and one is negative.

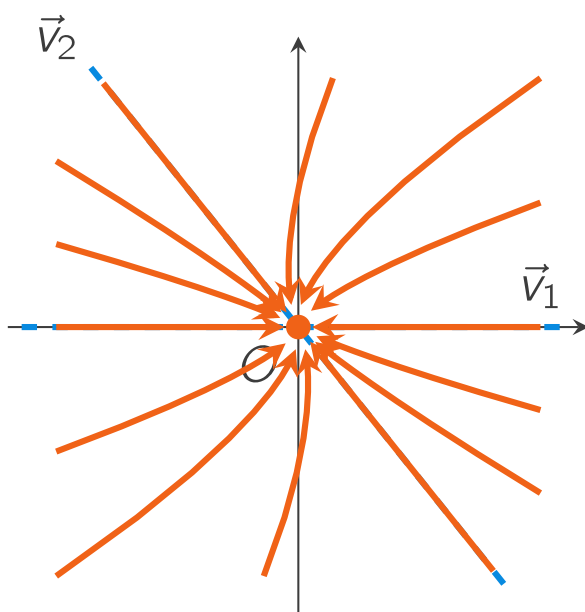
Of course, the eigenvalues of a  $2 \times 2$ -matrix can also be both negative or both positive. Then both factors  $e^{\lambda_1 t}$  and  $e^{\lambda_2 t}$  will either both decrease in time (when  $\lambda_1 < 0$  and  $\lambda_2 < 0$ ) or both increase in time, (when  $\lambda_1 > 0$  and  $\lambda_2 > 0$ ). The equilibrium points are then called *nodes*.

An equilibrium point  $\vec{X}_0$  is called a *saddle point* if the Jacobian matrix  $J(\vec{X}_0)$  has one negative and one positive eigenvalue. A saddle point is unstable because some of the solutions that start near the equilibrium point (here the origin) leave the neighborhood of the origin. A typical sketch of the solutions near a saddle point in the phase plane is given by



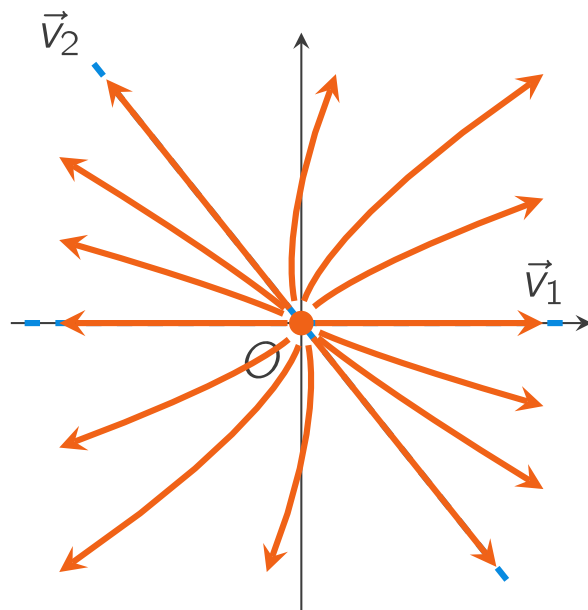
$$\lambda_1 < 0 \text{ and } \lambda_2 > 0$$

An equilibrium point  $\vec{X}_0$  is called a *stable node* if the Jacobian matrix  $J(\vec{X}_0)$  has two negative eigenvalues: all solutions that start near the equilibrium point stay near the equilibrium point. An example of a phase portrait is



$$\lambda_1 < 0 \text{ and } \lambda_2 < 0$$

An equilibrium point  $\vec{X}_0$  is called an *unstable node* if the Jacobian matrix  $J(\vec{X}_0)$  has two positive eigenvalues. A typical sketch of the solutions near an unstable node in the phase plane is given by



$$\lambda_1 > 0 \text{ and } \lambda_2 > 0$$

## Exercise A

2/2 points (graded)

Our system of differential equations for the rainbowfish and gouramis also has an equilibrium point at  $(0, 0)$ . Calculate the eigenvalues of the Jacobian matrix

$$J\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right),$$

and classify the equilibrium point.

The smallest eigenvalue  $\lambda_1$  equals

✓ Answer: -0.25

The largest eigenvalue  $\lambda_2$  equals

✓ Answer: 0.7

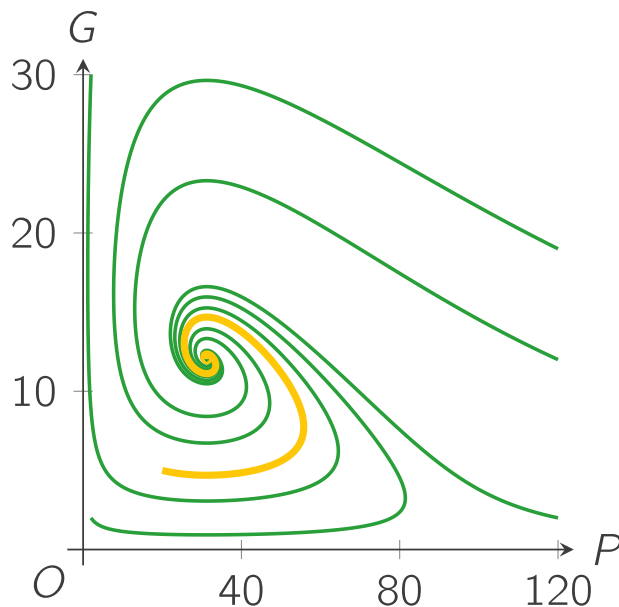
These eigenvalues indicate that equilibrium point  $(0, 0)$  is

- ☐ a stable node;
- ☒ a saddle point; ✓
- ☐ an unstable node;
- ☐ something else.

Submit

You have used 1 of 3 attempts

**i** Answers are displayed within the problem



Do you recognise the type of equilibrium point in  $(0, 0)$  which you calculated in the exercise?

On the next page, you will learn about even more types of equilibrium points.

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English ▼

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